Team Oil Spill MABLE Competition Agent

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ABSTRACT

Routing problems are part of a set of challenging problems that scale exponentially with the number of destinations. Parallel auctions between multiple agents are likewise difficult since success may depend on opponent preference behavior. The MABLE environment tasks agents to produce a strategy for auction bidding and route finding, combining these problems in a realistic oil-shipping simulator. We present our solution in the form of the Oil Spill agent. This agent combines a classic route finding algorithm with a classic auction strategy. We use a genetic algorithm with genome crossover and mutation to generate efficient trade schedules. We then calculate Shapely values to divide schedule cost among the individual trades available for auction.

KEYWORDS

Genetic Algorithms, Crossover, Shapely Values

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1 INTRODUCTION

Global maritime shipping is a massive industry, and an essential marketplace in the modern global economy. It is made up of many independent shipping companies, each competing to provide cheap, quick, or high throughput transportation of goods. In order to achieve efficient global maritime shipping, it is necessary for these companies to engage with a variety of challenges.

Routing problems are part of a set of challenging problems that scale exponentially with the number of destinations. Shipping companies consistently take on new contracts and must plan new routes through the ocean. This includes the usage of canals and international waters. Auctions between multiple agents are likewise difficult since success may depend on the behavior of other shipping companies. Companies that accurately predict the goals and opportunities for other companies as well as find a place for their own are more likely to find profits. They also face safety challenges, coordination challenges, and many, many others.

Some of these challenges are modeled within the MABLE cargo transportation simulator. This Python based environment allows agents to compete as shipping companies. Agents join reverse auctions to sell their services to market. Agents plan routes through Earth's oceans, rivers, and canals, to transport many goods to diverse destinations. Agents operate efficiently to decrease fuel cost and increase adaptivity. In these ways the simulation reflects some of the greatest challenges that shipping companies face today. Global shipping companies look to the field of Multiagent Systems to provide novel algorithmic solutions to these modern challenges.

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1.1 MABLE

The Maritime Shipping Competition 2025 provides an opportunity for Multiagent Systems researchers to present algorithmic solutions to maritime trading challenges.

The Maritime Agent-Based Logistics Emulator (MABLE) is a maritime trading simulator that allows for simulating company agents in shipping scenarios. It includes two major challenges: cargo bidding and transport scheduling.

1.2 Cargo Bidding

One of the central components of the MABLE simulator is the parallel auction. Company agents are invited to submit a bid for any or all of a set of trades offered each month. These trades are payments for the service of transporting cargo from a source port to a destination. The auction is a parallel reverse second-price auction. It is a reverse auction, so the sellers bid a price to provide their shipping service, and the service is won by the agent with the lowest offer. As a second-price auction, the winner is paid at the rate of the second-lowest offer. Since it is parallel, agents submit a bid for each trade before learning the result of the other trade offers. This poses a significant challenge to agents, who may be willing to offer low prices when a pair is received, but are less willing for a single trade, or vice versa. Bidding is repeated several times before the simulation terminates, giving agents a few shots at adaptation to opposing company agents.

1.3 Transport Scheduling

The other central component of the MABLE simulator is the vessel transportation scheduling. Company agents must deliver the cargo for the contracts they sold at the auction or pay a penalty. The contracts specify a location and time window for picking up the cargo, a destination location and time window, and a cargo size. The companies must schedule their heterogeneous fleet to perform efficient routing and minimize fuel cost across multiple rounds of bidding.

In this paper we present our strategy for the Maritime Shipping Competition 2025. In section 2, we present the foundational works that enabled our strategy, and related works. In section 3, we present the Oil Spill agent for strategy for the competition. In section 4, we conclude our discussion with future work in maritime transportation research.

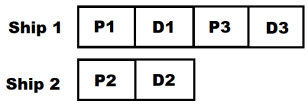


Figure 1: Parent 1

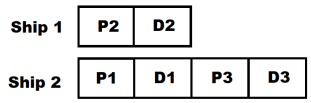


Figure 2: Parent 2

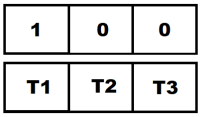


Figure 3: Trade Bit Map

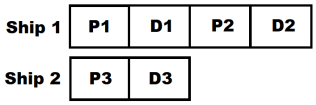


Figure 4: Child

2 RELATED LITERATURE

Our strategy uses genetic algorithms due to their effectiveness in routing optimization[2, 3]. In these algorithms, routes are represented by genetic codes that can be combined with other codes in a process called recombination. We are specifically relying on the order-based crossover technique[1].

We also use the permutation importance calculation capabilities of Shapely values[4, 5]. These have proven to be an essential measure of the value of a member within a coalition, and an excellent importance measure.

3 AGENT DESCRIPTION AND DISCUSSION

The agent utilizes a genetic algorithm to search for the route that maximizes the total number trades and minimizes the cost of the schedule. For the genetic algorithm, each schedule is the order in which the ports are visited by each ship to complete trades. The initial population is created by attempting to place a random trade into a random ship's schedule, and if the placement results in an invalid schedule, the trade is skipped. The genetic algorithm will always start with a population of valid solutions, but in the crossover and mutation, it will be possible for invalid schedules to appear. Overall, all the generations, the elite schedule is kept.

$$fitness = \frac{Trades * Cargo}{(Penalties + 1) * C}$$

The genetic algorithm uses an order-based crossover method. Each of the new trades in the auction are assigned a random bit map, which is mapped to each trade's pick up and drop off ports in the parent schedules. If a new trade is assigned a 1, the trade's pick up and drop off ports will be preserved in the child's schedule. If the new trade is assigned 0, then it's port's positions are replaced by unlocked trades from the opposite parent's schedule. Once a new schedule is created, any trades that appear in more than one ship are fixed so that the trade only appears once. The trade that remains on the same ship as it is in the dominant parents schedule is kept. An example of the crossover is shown in figures 1 through 4, and parent 1 is the dominant parent.

For mutation, the algorithm uses a simple operation that allows for the schedules to shrink and grow. From the list of new trades, a random trade is selected, which is removed from the scheduled. Then, with a probability of

$$\frac{1}{S+1}$$

, where S is the number of ships, the trade selected by chance will be placed randomly on the schedule.

Finally, the agent uses the Shapely value of each trade in the optimal schedule for its bidding. For bidding, the agent follows the philosophy of bidding the true cost of fulfilling a trade. The Shapely value allows the agent to accurately calculate how much each trade is contributing to the total cost of the schedule, and therefore, the Shapely value is the minimum the agent should bid for a given trade.

Table 1: Results of 9 trades per auction over 12 months

Agent	*Oil Spill*	*Arch Enemy*	*Scheduler*
Cost	18,393.652	23,227.937	49,634.058
Penalty	0	0	0
Revenue	20,759.302	19,197.143	35,921.585
Income	2,366.302	-4,030.795	-13,712.473

4 FUTURE WORK

There are several directions that could extend this strategy.

This strategy does not engage with opponent modeling. An adjustment to the fitness function that forms beliefs about opponent preferences could allow the agent to focus on the least demanded and most lucrative trades.

We have ongoing but incomplete research on considering the value of choosing to take a penalty for not filling out an expensive shipment, or for bidding on a shipment without the intention to fulfill the obligation. These strategies depend strongly on the punishment for failure. This strategy may also have negative legal consequences if employed by a real shipping company.

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